

Alperen Aydas

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SUMMARY

Multiplayer Systems Programmer and **Senior Game Developer** with over 4 years of professional game industry experience shipping competitive multiplayer titles. Expert in **client-server architecture**, **real-time systems**, and **performance tuning**. Demonstrated capability in **Native C++ networking**, including the development of custom authoritative network engines utilizing low-level transport protocols and data compression. Proven track record in managing full-cycle development, optimizing memory, and ensuring accurate state synchronization across distributed multiplayer systems.

SKILLS

Languages : C++, C#

Networking : ENet (UDP), Client-Side Prediction, Delta Compression, Server Reconciliation, Snapshot Interpolation, Lag Compensation

Engines : Custom C++ Engines, Unreal Engine 5, Unity 3D

Core Systems : Multithreading, Memory Optimization, Data-Oriented Design (ECS), Real-time performance tuning

PROJECTS

Purpose Network Engine | C++, ENet

- **Hybrid Architecture**: Built a dedicated **C++ Authoritative Server** communicating via a custom bit-packed, delta-compressed UDP protocol.
- **Zero-Allocation Architecture**: Engineered a memory-pooled network stack in C++, achieving a zero-allocation architecture that eliminates heap fragmentation during high-frequency real-time gameplay loops.
- **Competitive Netcode**: Implemented complex **client-side prediction** and **server reconciliation** to ensure responsive real-time movement, effectively mitigating 200ms+ latency and packet loss.
- **Lag Compensation**: Developed a **Server-Side Rewind** system with a 128-tick history buffer for **lag compensation** and hit-scan validation.
- **Scalability**: Tuned system performance and scalability by developing **Spatial Grid Interest Management (SGIM)** to cull entity updates, optimizing bandwidth for 1,000+ concurrent entities.

EXPERIENCE

Panteon

Jan. 2023 – Present

Senior Game Developer

Ankara, Turkey

- *Core Gameplay Programmer for "Rise of Heroes" (4X Strategy) and "Arcane Arena" (PvP Tower Defense)*
- **Arcane Arena (Live Ops)**: Implemented real-time multiplayer PvP features, collaborating closely with backend engineers to ensure **accurate state synchronization** and cheat prevention.
- **Rise of Heroes**: Owned the development of major social features including the **Alliance System** and **Chat Architecture**, managing complex asynchronous data states.
- Architected the **Scene Subsystem** to manage massive dynamic map object counts, achieving a **40% increase in FPS** through data-oriented optimization.
- Refactored legacy systems to support new design requirements during a technical pivot to a Vertical Survival RPG.
- Developed custom **Editor Tools** and content pipelines that significantly reduced iteration time for game designers.

Bytetyper

Sep. 2021 – Dec. 2022

Game Developer

Ankara, Turkey

- Managed the full-cycle development of **20+ prototypes**, collaborating directly with art and design teams.
- Successfully shipped 2 top-chart titles by optimizing workflows and ensuring reliable performance throughout the release cycle.

LEADERSHIP & VOLUNTEERING

Global Game Jam 2026 (ATOM)

Technical Mentor

Ankara, Turkey

- Selected by **ATOM** (Animation Technologies and Game Development Center) to mentor jam participants.
- Provided technical guidance on architecture, scope management, and debugging complex gameplay logic, demonstrating the strong interpersonal and problem-solving skills required for cross-discipline collaboration.

Game Jam Awards

- **ATOS (Magara Jam #3)**: 2nd Place / 600+ entries.
- **The Last Session (Magara Jam #5)**: 6th Place / 300+ entries.

EDUCATION

Hacettepe University

Sep. 2017 – June 2021

Bachelor of Science in Computer Engineering

Ankara, Turkey